Series And Exponential Functions

Erik An

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1 1.3 Series (Infinite Aggregation)

1.1 The Definition of Series

1.1.1 [Definition: Infinite Sums as Partial Aggregates]

1. Layer 1: The Logic & Spec View (QA Engineer)
   - **Concept:** An Infinite Series is not a static sum; it is a **Process**.
   - **The Object:** We take a sequence (a_n) (the inputs) and transform it into a sequence of **Partial Sums** (S_N) (the accumulator).
   - [cite$_{\text{start}}$]**The Check:** We test if the accumulator S_N converges to a limit S as $N \to \infty$[cite: 153, 155].
   - **State Tracking:**
     - 'Input[n]': a_n
     - [cite$_{\text{start}}$]'Accumulator[N]': $S_N = \sum_{n=0}^{N} a_n$ [cite: 152]
     - [cite$_{\text{start}}$]'Result': Limit of 'Accumulator'[cite: 155].
2. Layer 2: The Feynman View (Intuition)
   - **Stacking Bricks:**
     - Imagine stacking bricks of height \$a$_0$, a$_1$, a$_2$...\$.
     - S_N is the total height of the tower after N bricks.
     - "Convergence" means the tower stops growing and hits a specific ceiling height S, rather than growing infinitely tall.
3. Layer 3: Formal Source of Truth Given a sequence (a_n), we define the partial sums $S_N := \sum_{n=0}^{N} a_n$. If (S_N) converges, we write:

$$\sum_{n=0}^{\infty} a_n := \lim_{N\to\infty} \sum_{n=0}^{N} a_n$$

   [cite$_{\text{start}}$][cite: 152, 155]

1.2 Necessary Conditions & Geometric Series

1.2.1 [Lemma 1.3.1: The "Vanishing Terms" Sanity Check]

1. Layer 1: The Logic & Spec View (QA Engineer)

- **Goal:** Early rejection of divergent series.
- **The Rule:** If the input stream (a_n) does not go to zero, the sum **cannot** converge.
- **Logic Flow:**
  - 'IF' $\sum a_n$ converges 'THEN' $a_n \to 0$.
  - **Crucial Warning (Common Bug):** The reverse is **FALSE**. Just because $a_n \to 0$ does not mean the sum converges (e.g., Harmonic Series).
  - **Logic:** If you keep adding non-zero sized bricks, the tower grows forever.

2. Layer 2: The Feynman View (Intuition)

   - **The Leaky Faucet:**
     - If you are filling a bucket (S_N) with drops of water (a_n).
     - If the drops don't eventually become microscopic mist (approach 0), the bucket will overflow.
     - **Why it's not sufficient:** Even if the drops get smaller, if they don't get smaller **fast enough**, you still flood the room.

3. Layer 3: Formal Source of Truth If the sequence of partial sums (S_N) converges, then:

$$\lim_{n\to\infty} a_n = 0$$

   [cite$_{\text{start}}$][cite: 158]

1.2.2 [Theorem 1.3.2: The Geometric Series]

1. Layer 1: The Logic & Spec View (QA Engineer)

   - **Concept:** The "Base Case" for almost all convergence tests.
   - **Spec:** A series generated by a constant multiplier x.
   - **Condition:** Converges 'IFF' the multiplier's magnitude $|x|[cite_start] < 1$[cite: 168].
   - **Closed Form:** We have an exact formula for the sum, not just an approximation.

2. Layer 2: The Feynman View (Intuition)

   - **Zeno's Paradox:**

- Walk halfway to the wall (1/2), then half of what's left (1/4), then half again (1/8).
- You are adding infinite steps, but you never walk further than the wall (Sum = 1).
- If your step size multiplies by something ≥ 1, you walk forever.

3. Layer 3: Formal Source of Truth For $|x| < 1$:

$$\sum_{k=0}^{\infty} x^k = \frac{1}{1-x}$$

For partial sums ($x \neq 1$):

$$\sum_{k=0}^{N} x^k = \frac{1-x^{N+1}}{1-x}$$

[cite$_{\text{start}}$][cite: 165, 166]

1.3 Convergence Algorithms (Tests)

1.3.1 [Theorem 1.3.5: The Test Suite]

1. Layer 1: The Logic & Spec View (QA Engineer)

   - **Goal:** Determine system stability without calculating the exact final state.
   - **1. Comparison Test:**
     - Logic: If ‘UnknownSeries‘ $<$ ‘ConvergentSeries‘, then ‘Unknown‘ converges.
     - [cite$_{\text{start}}$]Logic: If ‘UnknownSeries‘ $>$ ‘DivergentSeries‘, then ‘Unknown‘ crashes (diverges)[cite: 188].
   - **2. Quotient (Ratio) Test:**
     - Logic: Compute the “decay rate” $\theta = |a_{n+1}/a_n|$.
     - [cite$_{\text{start}}$]Check: If $\theta < 1$, the series acts like a Geometric Series and converges[cite: 190].
     - Check: If $\theta > 1$, it diverges.
   - **3. Leibniz (Alternating) Criterion:**
     - [cite$_{\text{start}}$]Logic: If terms flip sign ($+/-$), get smaller, and hit 0, the oscillations dampen out and converge[cite: 193].

2. Layer 2: The Feynman View (Intuition)
   - **Comparison:** If I can fit my box inside your box, and your box fits in the room, my box fits in the room.
   - **Ratio Test:** "How fast is the signal dying?" If the signal strength drops by 50% every second ($\theta = 0.5$), it will eventually vanish.
   - **Leibniz:** A "dampening spring." You jump forward 1 meter, back 0.5m, forward 0.25m. You are zeroing in on a location.
3. Layer 3: Formal Source of Truth
   (a) **Comparison:** If $|a_n|[cite_start] \leq |b_n|$ and $\sum |b_n|$ converges $\implies \sum |a_n|$ converges[cite: 188].
   (b) **Quotient:** If $\exists \theta \in (0, 1)$ such that $|\frac{a_{n+1}}{a_n}|[cite_start] \leq \theta$, then $\sum |a_n|$ converges[cite: 190].
   (c) [cite$_{\text{start}}$]**Leibniz:** If $a_n \geq 0$ is decreasing and $a_n \to 0$, then $\sum(-1)^n a_n$ converges[cite: 193].

1.4 The Exponential Function

1.4.1 [Definition 1.3.11: The "Master" Function]

1. Layer 1: The Logic & Spec View (QA Engineer)
   - **Concept:** e^x is defined not by geometry, but by a specific power series.
   - **Stability:** Converges for **ALL** inputs x (Radius of convergence is ∞).
   - [cite$_{\text{start}}$]**Logic:** The $n!$ in the denominator grows so much faster than the power x^n that it crushes the term size to zero very quickly[cite: 231].
2. Layer 2: The Feynman View (Intuition)
   - **The Universal Translator:** This series turns **Addition** into **Multiplication** ($\exp(x + y) = \exp(x) \cdot \exp(y)$).
   - **Euler's Number:** e is just the output of this machine when you feed it 1.

3. Layer 3: Formal Source of Truth

$$\exp(x) := \sum_{n=0}^{\infty} \frac{x^n}{n!}$$

[cite$_{\text{start}}$][cite: 228] Functional Equation:

$$\exp(x + y) = \exp(x) \cdot \exp(y)$$

[cite$_{\text{start}}$][cite: 251]